METRIC
FF-B-171/1
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, NON-FILLING SLOT, ANSI/AFBMA DIMENSION SERIES 10, 02, 03 UNDER 10 MM BORE.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

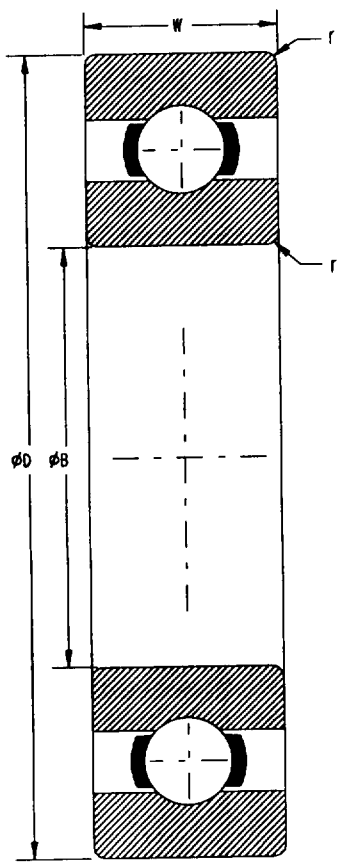
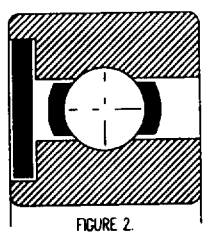
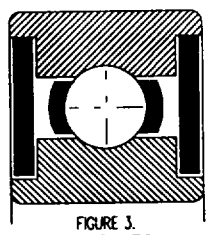
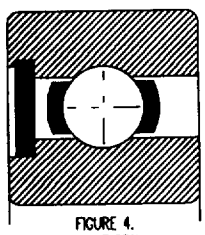
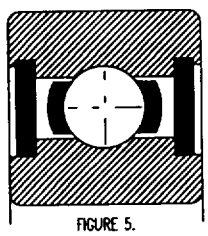


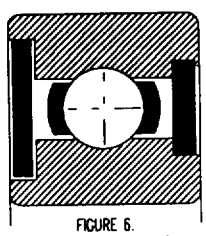
FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)


FIGURE 2.
SINGLE SHIELD
(CODE B)


FIGURE 3.
DOUBLE SHIELD
(CODE C)


FIGURE 4.
SINGLE SEAL
(CODE D)


FIGURE 5.
DOUBLE SEAL
(CODE E)


FIGURE 6.
SINGLE SHIELD AND SEAL
(CODE F)

BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and Seals. Shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Seals shall be impervious contact type. Shields and seals shall be specified in the part number by the codes in table III.

4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

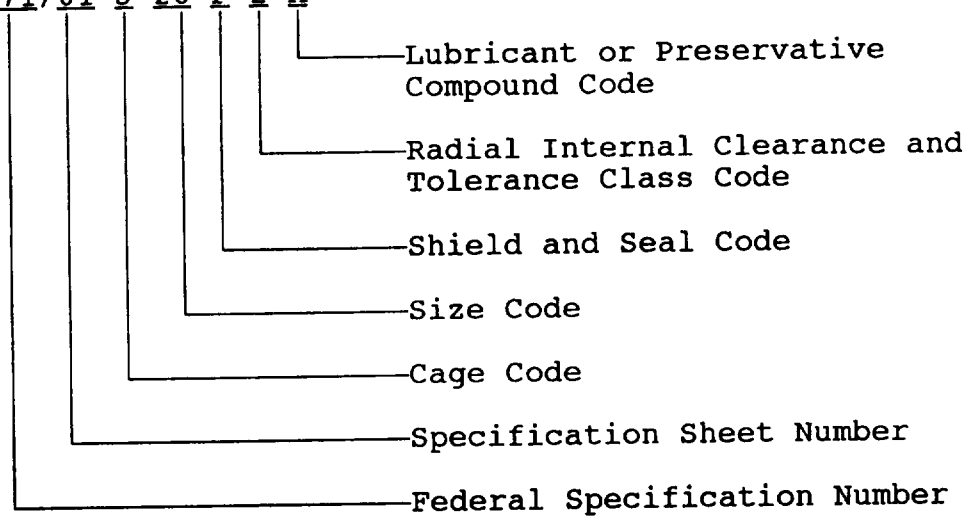
5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. Lubricant or Preservative Compound. Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the "C" suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the "A" or "B" suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

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7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and seals, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/01 J 26 F 2 A



FFB171/01J26F2A indicates - Bearing, cage-pressed steel or molded plastic; bore 6mm; outside diameter 19mm; width 6mm; one shield and one seal; radial internal clearance symbol 0 (normal); tolerance class ABEC-1; grease in accordance with MIL-G-81322.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder diameter Min 3/	C Dynamic load rating (pounds)
07	7	19	6	0.3	9	405
08	8	22	7	0.3	10	570
09	9	24	7	0.3	11	580
24	4	13	5	0.2	5.6	172
25	5	16	5	0.3	7	300
26	6	19	6	0.3	8	415
27	7	22	7	0.3	9	565
28	8	24	8	0.3	10	580
29	9	26	8	0.3	13	785
34	4	16	5	0.3	6	270
35	5	19	6	0.3	7	410

1/ Size codes for Table I were selected from dimension series 10 (dash numbers 07 thru 09) dimension series 02 (dash numbers 24 thru 29) dimension series 03 (dash numbers 34 and 35).

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

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TABLE III. Codes for Shields and Seals.

Code	Number of Shields and Seals.
A	none - fig.1
B	single shield - fig.2
C	double shield - fig.3
D	single seal - fig.4
E	double seal - fig.5
F	single shield and seal - fig.6

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

MILITARY INTERESTS:

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